
When One Modality Sabotages the Others: A Diagnostic Lens on Multimodal Reasoning

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Abstract

1 Despite rapid growth in multimodal foundation models, their reasoning traces
2 remain opaque: it is often unclear which modality drives a prediction, how conflicts
3 are resolved, or when one stream dominates. In this paper, we introduce *modality*
4 *sabotage*—a failure mode in which a high-confidence unimodal error overrides
5 other evidence and misleads the fused result. To analyze such dynamics, we propose
6 a lightweight, model-agnostic evaluation layer that treats each modality as an agent,
7 producing candidate labels and a brief self-assessment. A simple fusion mechanism
8 aggregates these outputs, exposing *contributors* (modalities supporting correct
9 outcomes) and *saboteurs* (modalities that mislead). Applying our diagnostic layer
10 in a case study on multimodal emotion recognition benchmarks with foundation
11 models revealed systematic reliability profiles, providing insight into whether
12 failures may arise from dataset artifacts or model limitations. More broadly,
13 our framework offers a diagnostic scaffold for multimodal reasoning, supporting
14 principled auditing of fusion dynamics and informing possible interventions.

15 1 Introduction

16 Multimodal foundation models have advanced rapidly in tasks that combine vision, language, and
17 audio, from question answering tasks to understanding social signals [1]. Yet in practice, their
18 decisions remain a black box: users cannot tell *which* stream of data the system relied on, *how*
19 conflicting evidence—e.g., when text, audio, and vision suggest different labels—was resolved, if
20 at all, or whether a single sensor dominated the outcome. Prior work has discussed related issues
21 such as *modality collapse*, where vision–language models over-rely on text [2], and *unimodal bias*,
22 where fusion lets one stream dominate across a dataset [3]. In contrast, we highlight a distinct
23 failure mode we call *modality sabotage*: instance-level cases where a high-confidence unimodal error
24 not only fails locally but actively overrides other evidence and pulls the fused prediction off-target.
25 Unlike collapse or bias, which describe systematic trends, sabotage is a diagnostic lens on individual
26 decisions, making visible which modality misled the model and when. Despite strong progress in
27 multimodal fusion [4–14] and impressive results from multimodal large language models (MLLMs)
28 in vision–language understanding, VQA, and video understanding [15–20], current systems mostly
29 emphasize cross-modal feature interaction and modality completion, leaving how cues map to
30 affective constructs and how conflicts are resolved largely unexplored. Decades of psychology and
31 affective computing show that audio and visual cues carry complementary emotional information [21–
32 23], for example facial expressions correlate with pleasant affect [24] while speech acoustics track
33 arousal [25–27]. Yet these studies typically isolate unimodal contributions rather than addressing
34 how models should integrate, arbitrate, or dominate across modalities in multimodal settings. We
35 address this need with a simple, transparent, model-agnostic framework that treats each modality as
36 an agent, whose outputs constitute a diagnostic layer that records per-modality votes, confidences, and
37 disagreements, enabling systematic analysis of contributions and failure modes before a final fused

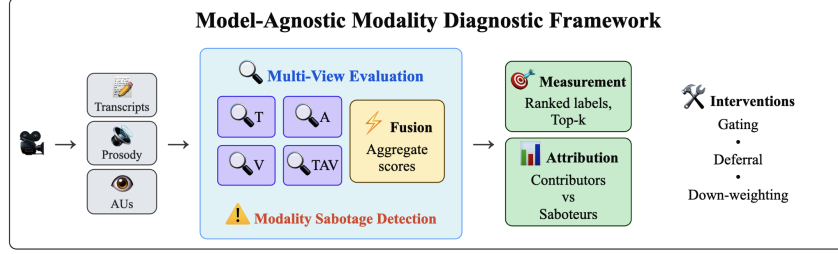


Figure 1: Each modality (T, A, V) and a joint view (TAV) agent outputs classification labels with confidence. A simple fusion aggregates these into a ranked prediction, enabling attribution of *contributors* vs. *saboteurs*. The callout highlights high-confidence unimodal errors that mislead the fused decision (*modality sabotage*); see Section 2 for details.

38 decision is made. Specifically, we propose a plug-and-play modality-as-agent fusion that queries text
 39 (T), audio (A), vision (V), and their joint view (TAV) separately, then aggregates their predictions into
 40 a final decision. The design makes attribution explicit at the instance level, surfacing *contributors*
 41 (modalities supporting correct answers) and *saboteurs* (modalities that mislead).

42 Our contributions are threefold: (i) a lightweight framework that yields instance-level attribution
 43 without retraining or architectural changes; (ii) a measurable operationalization of modality sabotage
 44 for high-confidence but misleading unimodal outputs; and (iii) dataset- and backbone-dependent
 45 reliability profiles that clarify whether failures stem from dataset artifacts or model limitations.

46 2 Methodology

47 We evaluate the framework across three public and widely-used multimodal emotion datasets and
 48 report unimodal and fused performance, top- k coverage, and sabotage diagnostics.

49 **Inputs per modality.** For each video segment we derive modality-specific, purely descriptive
 50 inputs that avoid direct emotion inference: (i) **Text (T):** Whisper ASR transcripts from the audio
 51 track serve as the textual input; (ii) **Audio (A):** Each audio utterance instance is analyzed by *Qwen-*
 52 *Audio* [28] with a structured prompt to elicit non-lexical descriptors—prosody (pitch/intonation,
 53 loudness/intensity, tempo/rhythm), voice quality (breathiness/creak/tension), and articulation—while
 54 forbidding use of lexical content or emotion labels; (iii) **Vision (V):** we compute facial AUs with
 55 *OpenFace* [29], select an AU-peak frame, and ask a VLM (*GPT-4 Vision* [30]) to produce an objective
 56 caption of observable cues (e.g. facial expressions, posture, gestures, and context) without mental-
 57 state attributions. These modality-specific descriptors feed the corresponding modality agents.

58 **Agents and outputs.** We propose a simple, model-agnostic framework (Figure 1) that treats
 59 each modality as an *agent* and makes fusion decisions legible. For each sample, T, A, V, and
 60 TAV are queried with a structured prompt; each agent returns a sorted set of candidate labels with
 61 confidence scores (1–100) together with a *data-quality report* (score, issues, and a short rationale).
 62 The confidences are sparse and uncalibrated (an agent may emit only a few labels), and the data-
 63 quality field is used to probe whether the LLM can self-diagnose issues such as noisy transcripts
 64 or occluded faces. We fuse agents by aggregating their confidences per label and normalizing to
 65 obtain a single ranked score vector. Let $S_m(y) \in [0, 100]$ be the confidence assigned by agent
 66 $m \in \{T, A, V, TAV\}$ to label y (zero if y is not proposed), and let $q_m \in [0, 1]$ denote the agent’s
 67 self-reported quality score (rescaled from 1–100). We compute

$$\tilde{s}(y) = \sum_m w_m S_m(y), \quad p(y) = \frac{\tilde{s}(y)}{\sum_{y'} \tilde{s}(y')},$$

68 where $w_m = 1$ by default and $w_m = q_m$ in a quality-weighted ablation. Across benchmarks, quality
 69 weighting did not improve top-1 accuracy (and sometimes reduced it), so we retain the unweighted
 70 variant as the main setting and report the weighted variant for completeness. We evaluate using the
 71 ranking induced by $p(y)$ and report top- k coverage.

72 **Modality sabotage (diagnostic).** Fusion can fail *silently* when an overconfident stream dominates:
 73 a wrong modality can pull the final decision off-target, and accuracy alone offers no attribution.

Let $S_m(y) \geq 0$ denote agent m 's evidence for label y (we use self-reported confidence 1–100; other signals such as probabilities or logits are interchangeable), $p_m(y) = S_m(y) / \sum_{y'} S_m(y')$, $y_m = \arg \max_y p_m(y)$, $c_m = \max_y p_m(y)$, and $\hat{y} = \arg \max_y \tilde{s}(y)$ with $\tilde{s}(y) = \sum_m w_m S_m(y)$. We distinguish two flavors: **Potential sabotage** for m holds when (i) $c_m \geq \tau$ (high confidence) and (ii) $y_m \neq y^*$ (its own error). **Successful sabotage** strengthens this by requiring (iii) $\hat{y} = y_m$ (the fused model follows m), with $\tau = 0.70$ unless noted. However, due to the nature of fusion, successful sabotage does not establish strict causality—multiple agents may jointly support the same wrong label. For this reason, in Section 3 we focus on **potential sabotage**, which provides a clearer upper bound on each modality's tendency toward overconfident errors. Both definitions nonetheless offer actionable diagnostic signals for gating, down-weighting, or deferral.

Top- k reasoning. Modality sabotage motivates $\text{Acc}@k$: in many sabotaged cases the correct label remains in the top- k list even when Top-1 is wrong, indicating recoverable uncertainty rather than pure failure. Concretely, we sort the fused scores $p(y)$ and report coverage at k ,

$$\text{Acc}@k = \frac{1}{N} \sum_{i=1}^N \mathbf{1}[y_i^* \in \text{TopK}(p_i)].$$

We report $\text{Acc}@k$ alongside Top-1 across datasets and backbones. These statistics do not alter the decision rule; they expose whether the diagnostic layer preserves sufficient signal under modality conflict to support reliable interventions, while being robust to the sparsity and lack of calibration in agent confidences.

3 Case Study Results

3.1 Aggregate accuracy and effect of self-reported quality

Table 1 compares a single-call **TAV** baseline (Top-1) with our **agentic fusion** (reported as “Fus T1–T5” in the table) under confidence-only fusion and reports the ablation when additionally weighting by self-reported data quality. Three patterns emerge. (i) *Top-1* vs. *Top- k* . The fusion maintains baseline-level Top-1 on **MELD** and **IEMOCAP** and improves markedly on MELD, while Top- k coverage rises steeply across datasets. On **MER**, Top-1 changes from 0.38 (baseline) to 0.33 (fusion, GPT-5-nano), but the correct label appears with high probability in the ranking (Top-5 = 0.97). On **MELD**, Top-1 improves by +0.09 for GPT-5-nano (0.27 $\rightarrow$ 0.36, +33%) and by +0.15 for GPT-4-mini (0.30 $\rightarrow$ 0.45), with Top-5 = 0.92/0.90. On **IEMOCAP**, Top-1 is essentially flat for GPT-5-nano (0.28 $\rightarrow$ 0.29) and slightly lower for GPT-4-mini (0.28 $\rightarrow$ 0.24), but Top-5 remains substantially higher than Top-1 (GPT-5-nano: 0.76, GPT-4-mini: 0.72). These results indicate that the fusion preserves multiple plausible hypotheses beyond the Top-1 prediction.

Ablation: confidence $\times$ quality weighting. The Δ columns quantify the change when scaling each vote by the product of its confidence and self-reported data-quality. Effects are small and often negative: e.g., on **MELD**/GPT-5-nano, $\Delta\text{Top-1} = -0.08$ and $\Delta\text{Top-2} = -0.06$; on **IEMOCAP**/GPT-5-nano, $\Delta\text{Top-1} = -0.05$ and $\Delta\text{Top-3} = -0.07$. Occasional mild gains appear (e.g., **MER**/GPT-4-mini: $\Delta\text{Top-4} = +0.02$, $\Delta\text{Top-5} = +0.03$), but the overall trend suggests that *self-reported quality* is a noisy proxy that does not consistently align with correctness. Consequently, we report confidence-only fusion as the main setting and treat quality weighting as an ablation.

3.2 Modality behavior and sabotage analysis

We operationalize *modality sabotage* as a measurable, instance-level diagnostic for *high-confidence, misleading* unimodal outputs that dominate the fusion and derail the final decision. This test makes the notion of “pulling the decision away” explicit, yields a *countable event* per example, and supports auditing by answering *who contributed* or *who hurt* each prediction. Figure 2 visualizes unimodal accuracy and sabotage rates per modality for GPT-5-nano under confidence-weighted fusion.

Diagnostic signals revealed. Beyond aggregate rates, the sabotage test surfaces actionable signals at the instance level as reported in Figure 2: (i) Across the columns, we observe a per-modality *calibration gap* (self-reported confidence vs. empirical accuracy), and (ii) Across the rows, we report the *dataset/backbone reliability profiles* ranking modalities by accuracy and sabotage. *Comparing the columns* in Figure 2 (Left), patterns are consistent: audio is the primary saboteur and text most

Table 1: **Top- k coverage and diagnostic effect of quality weighting.** The fusion maintains baseline-level Top-1 accuracy (“Fus T1” vs. “Base T1”) while substantially improving Top- k coverage (“Fus T2–T5”). The Δ block reports the change when switching from *confidence-only weighting* to *confidence $\times$ data-quality weighting*. Comparisons across datasets and backbones (GPT-5-nano vs. GPT-4o-mini) highlight systematic differences in modality reliability and pipeline robustness.

Dataset / Model	Accuracy						Δ (confidence+quality vs. confidence-only)				
	Base T1	Fus T1	Fus T2	Fus T3	Fus T4	Fus T5	Δ T1	Δ T2	Δ T3	Δ T4	Δ T5
MER / GPT-5-nano	0.38	0.33	0.62	0.85	0.92	0.97	+0.00	+0.01	+0.00	-0.02	+0.01
MER / GPT-4o-mini	0.35	0.23	0.52	0.75	0.83	0.85	-0.03	+0.00	+0.00	+0.02	+0.03
MELD / GPT-5-nano	0.27	0.36	0.58	0.73	0.86	0.92	-0.08	-0.06	-0.03	-0.03	-0.04
MELD / GPT-4o-mini	0.30	0.45	0.64	0.76	0.85	0.90	-0.02	+0.01	-0.02	-0.02	-0.02
IEMOCAP / GPT-5-nano	0.28	0.29	0.47	0.62	0.73	0.76	-0.05	-0.07	-0.07	-0.02	+0.03
IEMOCAP / GPT-4o-mini	0.28	0.24	0.43	0.60	0.70	0.72	+0.01	+0.03	+0.00	-0.02	+0.00

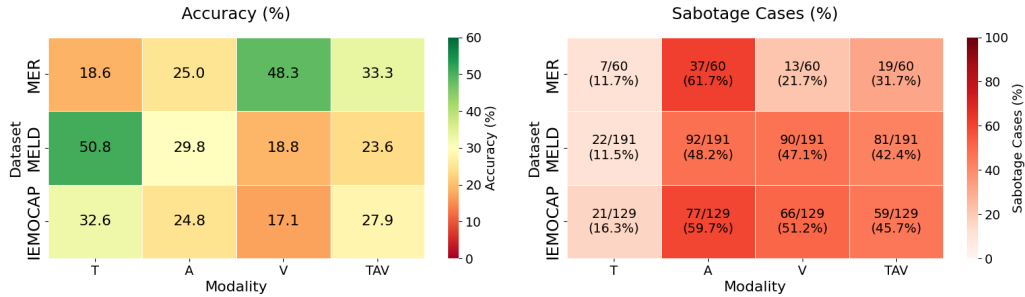


Figure 2: *Left heatmap*: unimodal accuracy for Text (T), Audio (A), Vision (V), and joint view (TAV), highlighting differences across datasets. *Right heatmap*: proportion of cases where a modality *sabotages* the fused decision (high-confidence error flipping Top-1 at threshold 70), where each values show #cases/total (rate%).

contributed. This provides a basis for identifying which components of a model pipeline may require refinement. Across rows, we can evaluate which modalities are less reliable within each dataset. This is consistent with each dataset characteristics: **MER** suffers from noisy ASR/translation but benefits from rich video cues; **MELD**’s sitcom-style video with exaggerated cues or multiple actors can mislead vision; **IEMOCAP** features seated dyads, where acted expressions and experimental scenes limit visual reliability.

Proposed Uses of Modality Sabotage Diagnostics Once modality sabotage events are identified, their associated confidence patterns can be leveraged not only for auditing but also for improving reliability. For example, sabotage scores can guide retraining with harder negatives, reweighting modalities that are systematically overconfident, or gating streams that frequently sabotage decisions. Beyond training-time interventions, sabotage scores can be incorporated directly at inference as weights on the final decision layer, adjusting the aggregated ranking to down-weight suspect modalities before producing the output. In this way, the diagnostic layer becomes actionable, informing both model development and real-time decision policies.

4 Conclusions

We introduced a lightweight, model-agnostic modality-as-agent framework that makes multimodal fusion decisions interpretable at the instance level. Each modality produces candidate labels with confidences, and a simple fusion aggregates them into a ranked prediction, making visible both supportive and misleading influences. By operationalizing modality sabotage as a measurable diagnostic, the framework provides concrete signals for auditing fusion dynamics, detecting overconfident errors, and guiding interventions such as gating or down-weighting unreliable streams. In this paper, we demonstrated the feasibility of the approach on multimodal emotion recognition. Looking forward, we position this diagnostic as a scaffold for multimodal algorithmic reasoning and broader agentic AI, offering a lightweight tool to expose shortcomings, handle unreliable sources, and build more interpretable multimodal systems.

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